MAGNESIUM SULFATE

SEIZURE PROTECTION / assess breathing, reflexes + urine output



WHAT IT DOES

Prevents/treats seizures in **severe preeclampsia / eclampsia**.
Not the BP medication. [1]

WHY KIDNEYS MATTER

Magnesium is cleared by the kidneys. Low urine output or renal impairment increases accumulation and toxicity risk. [2]

01 MONITOR THE WHOLE PATIENT

[1-3]

BREATHING

Respiratory rate + depth,
oxygen saturation,
work of breathing.
Watch for depression.

REFLEXES + LOC

Deep tendon reflexes,
alertness, weakness.
New absent reflexes are concerning.

URINE + FLUIDS

Strict intake/output.
Trend renal function.
Low output raises toxicity risk.

ALSO CHECK
BP/pulse, infusion pump and line, ordered magnesium levels, and fetal status. Use the unit's assessment frequency and hold parameters. [1-3]

02 TOXICITY: RECOGNIZE → RESPOND

[1-3]

CONCERNING FINDINGS

Slow/shallow breathing or falling oxygen
Absent reflexes
Marked drowsiness / weakness
Low urine output or worsening renal function

ACT PROMPTLY

Stop the magnesium infusion.
Call the obstetric / emergency team.
Support airway and breathing.
Obtain ordered levels and reassess.

ANTIDOTE: CALCIUM GLUCONATE

Keep immediately available. Give IV **as ordered / per emergency protocol**.
Do not wait for a lab result when respiratory compromise is developing.

EXPECTED ≠ TOXICITY

Warmth or flushing may occur.
Respiratory depression and lost reflexes are not normal reassurance.

EXAM SEPARATION

Magnesium: seizure prevention/treatment.
Labetalol / hydralazine / nifedipine: ordered BP control.

SOURCES | Click a title; full links are in the companion text file.

[1] ACOG Practice Bulletin 222: Gestational Hypertension and Preeclampsia (2020)
[2] DailyMed: Magnesium Sulfate in Dextrose - prescribing information (2026)
[3] Safer Care Victoria: Hypertension in Pregnancy (2024)

Prepared October 21, 2026 | Nursing study aid. Follow ordered doses, monitoring and emergency protocols.